Behavioral Run-off of Non-Maturing Deposits

Theory and Method of the DAV Model (IRRBB, pure-stdlib)

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Contents

Preface

Theory companion to the DAV run-off model (`BNP_Work/runoff_model/`). Goal: derive the run-off curve $S(t)$ for non-maturing demand deposits (DAV, Algeria) for IRRBB, in pure Python stdlib. Dense by design: definitions, equations, results. The code is the *how*; this is the *why*.

Two binding constraints. (i) **Stdlib only** — no `numpy`/`pandas`/`sklearn`/`matplotlib`; every estimator is hand-rolled. (ii) **Time-scarce** — $N_{\text{acct}} \gg T$, with $T \approx 120$ months and few macro cycles, so honest uncertainty is *temporal* and a high-capacity model overfits out-of-time. Both recur throughout.

Notation

Symbol	Meaning
$i,\ t$	account index; monthly (event-)time index
$\tau,\ \mathcal{F}_\tau$	decision time; information knowable at τ
T_i	event month for i (closure / dormancy / balance < floor)
$h_i(t)$	hazard $= \mathbb{P}(T_i = t{+}1 \mid T_i > t,\ \mathcal{F}_t)$
$S_i(t) = A_i(t)$	account survival $= \mathbb{P}(T_i > t)$; A when contrasting with erosion
$r_i(t)$	retention given alive $= B_i(t)/B_i(0) \mid$ survival
$B_i(t) = B_i(0)A_i(t)r_i(t)$	account balance run-off
$S(t),\ B(t)$	book curves, balance-weighted aggregates
$B_i(0)$	as-of balance (run-off weight), `CTRVL KDA`, KDA
$\mathbf{z}_{i,t}$	causal feature vector, $\mathcal{F}_t$-measurable
$\boldsymbol{\beta},\ b_0,\ \sigma$	hazard coefs, intercept, sigmoid $\sigma(x) = (1 + e^{-x})^{-1}$
$\mathrm{Sig}^N(X)$	depth-N truncated signature
$Q,\ p_0$	CTMC generator, initial regime law
H	max evaluated horizon (months); sets purge/embargo
WAL	$= \sum_{t\ge 0} S(t)\,\Delta t = \sum_{t\ge 1}(S(t{-}1) - S(t))\,t$

Conventions

- **Causality.** Anything feeding a prediction at τ uses data with availability $\leq \tau$; fitted statistics use a required `fit_end`, applied forward.
- **Calibration $\gg$ discrimination.** $S(t) = \prod(1 - h)$ amplifies hazard-*level* bias, not ranking error (Ch. 3, 11).
- **Honesty.** All current numbers are methodology results on synthetic real-format data; the real panel is on the bank PC. Fit-only / scenario-only / not-OOS-validatable claims are flagged.
- Balances in KDA; aggregates are balance-weighted (whale-dominated book).

Chapter 1

The Problem

1.1 Setup

A demand deposit (DAV) is contractually repayable on demand ($T_{\text{contract}} = 0$) yet the *book* balance is sticky. IRRBB needs a behavioral maturity replacing the missing contractual one.

Definition 1.1 (run-off curve). Normalize today's stock to 1. $S(t)$ = expected fraction still present at horizon t; $S(0) = 1$, S weakly decreasing. $1 - S(t)$ = cumulative run-off.

Summaries used downstream:

$$\text{WAL} = \sum_{t \geq 0} S(t)\, \Delta t, \qquad D_{\text{eff}} = -\frac{1}{S} \frac{\partial S}{\partial y}. \tag{1.1}$$

Uses: IRRBB (ΔEVE/ΔNII bucketing under rate shocks), FTP (tenor crediting).

1.2 Decomposition

Three behaviors drive the curve: **attrition** (discrete event, whole balance leaves), **erosion** (continuous drawdown of a surviving balance), **core** (inert sticky float $\to$ long tail). Make it explicit per account:

$$B_i(t) = B_i(0)\ \underbrace{A_i(t)}_{\text{alive?}}\ \underbrace{r_i(t)}_{\text{kept}|\text{alive}}\ , \qquad A_i(t) \in [0,1],\ r_i(0) = 1. \tag{1.2}$$

Book aggregates (balance-weighted, because the book is whale-dominated):

$$B(t) = \frac{\sum_i B_i(0) A_i(t) r_i(t)}{\sum_i B_i(0)}, \qquad S(t) = \frac{\sum_i B_i(0) A_i(t)}{\sum_i B_i(0)}. \tag{1.3}$$

$S(t)$ = attrition-only sub-curve (Ch. 3–5); $B(t)$ = deployed curve, adds erosion r (Ch. 6).

Remark 1.1. A raw weighted-average of historical balances = the convention/ECM floor (Ch. 7); it discards seasoning, segment, regime, balance-bucket heterogeneity. The behavioral model (1.2) keeps it and aggregates *modeled* survival — a bet that must win on an out-of-time backtest (Ch. 11), not be asserted.

1.3 Three structural difficulties

1. **Forward extrapolation, short data.** $S(t)$ predicts the future decay of today's stock $\rightarrow$ out-of-time is the only honest test. With $T \approx 120$ and few cycles, the curve beyond ~ 24 months is *structural extrapolation*, not validated.
2. **Uncertainty is temporal.** Accounts are many but share each month's macro shock $\rightarrow$ not independent. Honest bands resample *time*, not accounts (Ch. 10).
3. **Gate A (rate identifiability).** A rate elasticity needs rate variation. The Algerian policy rate is administered (3 distinct values, 92-month flat run) $\rightarrow$ not identifiable; substitute the money-market rate, lean stress on inflation + scenarios, disclose. Run *before* any rate-conditional fit.

These three dictate the design: small, regularized, time-block-validated, explicit about measured-vs-assumed.

Chapter 2

The Data: Survival Panel

The model fits a flat table; survival structure and no-look-ahead live in the **data layer**.

2.1 Panel construction

Input: monthly dumps `DAV_MMYYYY.txt` (filename = authoritative period). Hazards: title preamble, UTF-16/cp1252, US dates, comma-thousands, **column drift** → resolve columns *by header name*. Output: account-month rows (i, t) with `CTRVL KDA`, type, `DATE OUVERTURE`, `segment`.

- Keep all deposit products; tag `segment` $\in$ {vue_dinars, vue_bu, garantie}. Guarantee deposits run off on contract lifecycle → **separate hazard segment**, never pooled, never dropped.
- **Integrity gate:** $\sum_i$ `CTRVL KDA` per month must reconcile to the known book total. Non-reconciling panel ⇒ wrong population; nothing downstream fixes it.

2.2 Survival bookkeeping

Definition 2.1 (event). T_i = first month i disappears (closure/dormancy) **or** balance < floor (economic withdrawal). Floor is a degree of freedom → stress-tested (Ch. 11).

Two finite-window effects:

- **Right-censoring.** Alive at window end ⇒ $T_i > T_{\text{end}}$ only. Keep the account — it is the stickiest balance; it contributes a run of $y = 0$ rows then stops (handled natively by the discrete likelihood, Ch. 3).
- **Definition 2.2 (left-truncation).** Opened before the window ⇒ enter the risk set at T_{start} with seasoning > 0, likelihood **conditioned on survival to entry**; tenure measured from `DATE OUVERTURE`, not from T_{start}.

Output = a **risk set**: per month, accounts under observation, each tagged alive/event, correct tenure clock.

2.3 Event-time / point-in-time (PIT)

A flat table has no clock; inject it as data. Three obligations (panel builder):

1. **Elapsed time as columns:** seasoning, calendar month/year, $\Delta\tau$ since last event.
2. **PIT as-of macro join.** Each macro row carries `available_date`. Attach to (i, τ) the latest macro obs with available_date $\leq \tau$ (backward as-of: sort + `bisect`). Never the reference-month or revised vintage.
3. **No fitted statistic peeks.** Scalers, bucket cuts, λ, regime params, signature standardizers all fit on train, applied forward. `fit_end` is required, not default (the Step-3 leak).

Remark 2.1. The model is time-agnostic; event-time handling = (i) panel as-of join + elapsed-time columns, (ii) splitter time-predicates (Ch. 11). Nothing in the estimator.

Design fork. (A, default) features irregular/PIT, **decision grid regular monthly** $\to S(t) = \prod(1-h)$ coherent, regulatory horizons legible. (B) fully continuous-time hazard (Cox + exposure) — more correct only if irregular-horizon output is required; rejected (data are monthly, harder in stdlib).

2.4 Gate A (real-data result)

Policy rate administered (3 values, 92-mo flat) $\to$ use **money-market rate** (118 distinct, std $\approx$ 0.91). Inflation 0.1–10.8% = solid driver. Frozen OOS 2023–24 = single stagnant regime $\to$ regime-stress $S(t)$ is **fit-only, not OOS-validatable**. Resolved here, before fitting.

Chapter 3

Discrete-Time Survival and the Hazard

Cast attrition as discrete-time survival $\Rightarrow$ it *becomes* penalized logistic regression on person-month rows.

3.1 Hazard and survival

Definition 3.1. $h_i(t) = \mathbb{P}(T_i = t{+}1 \mid T_i > t,\ \mathcal{F}_t)$, $S_i(t) = \mathbb{P}(T_i > t \mid \mathcal{F}_0)$.

Proposition 3.1. $S_i(t) = \prod_{k=1}^{t} (1 - h_i(k))$. (3.1)

Proof. $\mathbb{P}(T_i > t) = \prod_{k=1}^{t} \mathbb{P}(T_i > k \mid T_i > k{-}1)$, each factor $= 1 - h_i(k)$. $\square$

(3.1) is the engine: model the 1-month hazard, chain to the whole curve (`survival_path`: `cur*=(1-h)`).

3.2 Likelihood

Per at-risk month, $y_{i,t} = \mathbf{1}\{T_i = t{+}1\}$. Risk set $\mathcal{R}_i$ = observed at-risk months.

$$\mathcal{L} = \prod_i \prod_{t \in \mathcal{R}_i} h_i(t)^{y_{i,t}} \big(1 - h_i(t)\big)^{1 - y_{i,t}}. \tag{3.2}$$

This is exactly a **pooled Bernoulli (logistic) regression** on person-month rows. Censoring and truncation need no extra machinery:

- right-censored $\Rightarrow$ a string of $y = 0$ rows, future absent from (3.2);
- left-truncated $\Rightarrow$ enters $\mathcal{R}_i$ from observation start $\Rightarrow$ auto-conditioned on survival to entry.

3.3 Parameterization

$$h_i(t) = \sigma(b_0 + \boldsymbol{\beta}^\top \mathbf{z}_{i,t}), \qquad \mathbf{z}_{i,t}\ \mathcal{F}_t\text{-meas.} \tag{3.3}$$

Design:

- **Pooled** (one $\boldsymbol{\beta}$, heterogeneity via $\mathbf{z}_{i,t}$): per-account overfits (rare events, short histories); raw-aggregate discards cross-section (= ECM, Ch. 7).
- **Balance-weighted loss** $w_{i,t} \propto B_i(0)$:

$$\min_{b_0,\boldsymbol{\beta}}\ -\sum_{i,t} w_{i,t}\big[y_{i,t}\log h_i + (1-y_{i,t})\log(1-h_i)\big] + \lambda\,\Omega_\alpha(\boldsymbol{\beta}). \tag{3.4}$$

  Book is whale-dominated ⇒ weight by balance, else the fit optimizes the typical *small* account. Ω_α = elastic net (Ch. 4).

- **Competing risks (option):** multinomial/softmax cause-specific $h_i^{(c)}$, aggregated to stock decay. Default: single event.

3.4 Aggregation

For accounts alive at the as-of date, roll (3.1) to $S_i(t) = A_i(t)$, then

$$S(t) = \frac{\sum_i B_i(0)A_i(t)}{\sum_i B_i(0)}, \quad t = 0, \dots, H. \tag{3.5}$$

$\times r_i(t) \Rightarrow B(t)$ (Ch. 6). Stressed curves = re-score with shocked macro (Ch. 9).

3.5 Metric: calibration, not discrimination

$S(t) = \prod(1-h) \Rightarrow$ a product is sensitive to systematic bias in each h, insensitive to the ranking of the h's. So judge *levels*:

- **reliability** (predicted h vs realized frequency by bin, also balance-weighted);
- **PIT** (randomized PIT residuals $\approx$ Uniform$(0,1)$, KS test);
- **Brier** (proper scoring rule; AUC is *not*, it rewards ranking only).

Synthetic deployed hazard: OOT ECE ≈ 0.003, PIT-KS ≈ 0.017. Headline of the validation file (Ch. 11).

Chapter 4

Penalized Logistic Readout

Hundreds of collinear features (signatures, Ch. 5) on $T \approx 120 \Rightarrow$ unpenalized fit overfits and is unstable. Elastic net makes (3.4) well-posed.

4.1 Penalty

$$\Omega_\alpha(\boldsymbol{\beta}) = \alpha\|\boldsymbol{\beta}\|_1 + \tfrac{1}{2}(1-\alpha)\|\boldsymbol{\beta}\|_2^2, \qquad (\ell_1, \ell_2) = (\lambda\alpha,\ \lambda(1-\alpha)), \quad b_0 \text{ unpenalized.} \tag{4.1}$$

- **L1** $\rightarrow$ exact zeros (selection); needed since most signature terms are noise.
- **L2** $\rightarrow$ shares weight across collinear groups (signatures are shuffle-product collinear); stabilizes what pure Lasso (α=1) makes unstable.
- Pure Ridge (α=0) never selects. Tune (λ, α) out-of-time (Ch. 11). Standardize features internally; map coefs back to raw space.

4.2 Solver: coordinate descent (glmnet)

IRLS quadratic approx of the logistic loss, then cyclic CD. Per coordinate, closed form via the soft-threshold $\mathcal{S}_\kappa(z) = \operatorname{sign}(z)\max(|z| - \kappa, 0)$:

Proposition 4.1 (CD update).

$$\beta_j \leftarrow \frac{\mathcal{S}_{\ell_1}\big(\sum_i \tilde{w}_i z_{ij} r_i^{(-j)}\big)}{\sum_i \tilde{w}_i z_{ij}^2 + \ell_2}, \tag{4.2}$$

$r_i^{(-j)}$ = partial residual excl. j, $\tilde{w}_i$ = IRLS weights. Soft-threshold = L1 selection; $+\ell_2$ = ridge shrinkage. Closed-form, no step tuning, stable on collinear features. `solver='pgd'` (ISTA: grad step + prox) gives identical coefs (cross-check).

Remark 4.1. Not used: *projected* GD (that solves the constrained $\|\beta\|_1 \leq t$ form); **Huber** (Bernoulli has no outliers) — Huber belongs on the continuous ECM/erosion regressions (Ch. 6–7).

4.3 Overfit control

1-SE rule (default). Let $\mathrm{CV}(\lambda)$ = mean OOT score over walk-forward origins, SE = its s.e. across origins. Deploy

$$\lambda^\star = \max\{\lambda : \ \mathrm{CV}(\lambda) \le \mathrm{CV}(\lambda_{\min}) + \mathrm{SE}(\lambda_{\min})\}. \tag{4.3}$$

Among statistically-indistinguishable models, pick the sparsest. On the (flat) synthetic HP surface the raw argmin chases noise; 1-SE picks the defensible model.

No early stopping. The penalized logistic is *convex*; CD converges *to* the optimum of (4.1). Stopping early = a worse solve of the same problem, not a milder regularizer. So `lr`, `epochs`, `max_irls` are convergence settings, fixed. Real overfit controls: λ (1-SE), purged CPCV, frozen OOS tail, conformal coverage (Ch. 10–11).

4.4 HP search

Only (λ, α). **λ-path** $\times$ **α-grid**, scored by walk-forward OOT **NLL/Brier** (proper scoring, *not* AUC). Grid (not Optuna/TPE) = deterministic, auditable for governance. Parallel via stdlib `multiprocessing` ($\sim$ 6–7$\times$ on 20 cores, identical to serial). The grid sets L1 strength $\Rightarrow$ feature inclusion is never hand-tuned; we only count surviving terms.

Chapter 5

Path Signatures (Nonlinearity)

(3.3) is linear in $\mathbf{z}$. Path *shape* may carry hazard signal. Stdlib, thin data $\Rightarrow$ use a fixed nonlinear feature map (signature) + the linear readout (Ch. 4), not a learned net.

5.1 Definition

Definition 5.1. For a path $X : [0,T] \to \mathbb{R}^d$, the signature is the family of iterated integrals

$$\mathrm{Sig}(X) = (1, (S^i)_i, (S^{ij})_{i,j}, \dots), \qquad S^{i_1\cdots i_k} = \int_{0<t_1<\cdots<t_k<T} dX^{i_1}_{t_1} \cdots dX^{i_k}_{t_k}. \tag{5.1}$$

Truncate at depth N: Sig^N. Level k = order-k interactions of increments (depth 1: net increment; depth 2: signed areas / covariations; …). Dimension

$$\dim \mathrm{Sig}^N = \frac{d^{N+1}-1}{d-1} \quad \text{(exponential in } N\text{)}. \tag{5.2}$$

This blow-up makes the Ch. 4 penalty mandatory. Compute by iterated *sums* ($\sim$ 50 lines); extend in time by **Chen's identity** (signature of concatenation = tensor product) $\Rightarrow$ causal per-τ updates are cheap.

5.2 Mandatory pre-transforms

Raw signature is reparametrization-invariant and drops level/absolute-time/quadratic-variation — exactly what we need. Break the invariances:

- **time augmentation:** append channel τ (and seasoning) $\Rightarrow$ absolute time = continuous positional encoding;
- **basepoint:** prepend start value $\Rightarrow$ absolute level;
- **lead-lag:** embed vs 1-lag copy $\Rightarrow$ depth-2 terms capture quadratic variation (volatility).

Channels: log(balance), money-market rate, inflation, (seasoning). Standardize fold-local; sweep depth $N \in \{2, 3\}$.

5.3 The trade (honest)

Signature = *fixed* features + *linear* weights; a net learns both. More rigid — an asset here, not superiority.

- **Lost:** learned adaptive representation; selective attention (a signature *summarizes* the path, cannot spotlight one event).
- **Kept:** depth-2 cross-channel interactions; positional info (τ, seasoning channels); interaction-order control (depth); variable-length $\rightarrow$ fixed-vector.
- **Costs:** hard truncation past N; exponential dimension ($\Rightarrow$ penalty); invariances must be engineered back; worse interpretability (ablation/PDP mitigate, Ch. 8/11).
- **Why loss is small here:** $T \approx 120$ cannot fund a learned high-capacity model OOT; bottleneck is data/signal, not architecture. A sharp pattern above depth N and below noise floor is unverifiable on 120 months $\Rightarrow$ claiming a net would catch it violates the honesty rule.

5.4 Empirical verdict (synthetic)

Tested through the production selector (grid tunes L1):

$$\text{raw (4)} = 0.1710, \quad \text{raw+oracle-path (5)} = 0.1710 \text{ (tie)}, \quad \text{raw+sig (46)} = 0.1720 \ (-0.6\%).$$

Key: the **oracle** one-step balance-drop feature also ties raw $\Rightarrow$ current log(balance) already absorbs path info $\Rightarrow$ the synthetic is Markovian-enough $\Rightarrow$ test is **uninformative, not negative**. Signatures = auto-re-tested candidate, default off, decided on the real panel (Ch. 11).

Chapter 6

Balance Erosion and Combined Run-off

Hazard gives $A_i(t)$. For DAV, surviving accounts also draw down (erosion $r_i(t)$); ignoring it understates run-off.

6.1 Two channels

$B_i(t) = B_i(0)A_i(t)r_i(t)$:

- $A_i(t) \in [0, 1]$ — survival probability (discrete event), the hazard;
- $r_i(t) = B_i(t)/B_i(0) \mid \text{alive}$ — continuous shrinkage, $r_i(0) = 1$.

Modeled separately (different statistics, different drivers, desk wants them apart).

6.2 Retention model

Log scale (positivity, multiplicative → additive). Target $\Delta \log B_{i,t} = \log B_{i,t} - \log B_{i,t-1}$, **linear elastic net** on the causal features; reconstruct

$$r_i(t) = \exp\Big(\sum_{k=1}^{t} \widehat{\Delta \log} B_{i,k}\Big). \tag{6.1}$$

Same PIT / walk-forward / balance-weighting.

Remark 6.1 (Huber here). $\Delta \log B$ has genuine outliers (one-off transfers) ⇒ **Huber loss** ($\sim 8\times$ more robust at 5% contamination). The binary hazard cannot have outliers, so Huber is for this continuous regression, not Ch. 3.

6.3 Combined curve

$$B(t) = \frac{\sum_i B_i(0) A_i(t) r_i(t)}{\sum_i B_i(0)}, \quad A(t) = \frac{\sum_i B_i(0) A_i(t)}{\sum_i B_i(0)}, \quad r(t) = \frac{\sum_i B_i(0) r_i(t)}{\sum_i B_i(0)}. \tag{6.2}$$

Gap $A(t) - B(t)$ = erosion contribution. Synthetic (declining gen): $B(12) \approx 0.89$ vs $A(12) \approx 0.97$ $\Rightarrow$ most year-1 run-off is drawdown, not closure.

Pitfall. Upward balance drift $\Rightarrow r(t) > 1 \Rightarrow$ accretive "run-off" > 1 (nonsense). Synthetic gen flipped to -0.4%/mo to fix. On real data the sign is empirical (high-inflation nominal growth can give $r > 1$ legitimately — separate IRRBB decision).

Deployed quantity $= B(t)$; $S(t) = A(t)$ is the attrition sub-component. Daily scorer dispatches on the selected run-off model (Ch. 7).

Chapter 7

Baselines: Convention and ECM

The hazard must beat these out-of-time, else ship the simpler one (governed selection, Ch. 11).

7.1 Convention (core/volatile)

Definition 7.1. Over a trailing window: core = trailing min (or low quantile) of book balance; volatile = excess. Amortize each linearly:

$$B_{\text{conv}}(t) = \text{core}\left(1 - \frac{t}{L_{\text{core}}}\right)_+ + \text{vol}\left(1 - \frac{t}{L_{\text{vol}}}\right)_+, \quad L_{\text{core}} \approx 60,\ L_{\text{vol}} \approx 3. \tag{7.1}$$

Transparent, no estimation; the floor. Static $\Rightarrow$ no conditioning, no sensitivity.

7.2 ECM (error-correction)

Cointegration: $\log D_t$ and macro share a stationary long-run relation $\log D_t^* = \gamma_0 + \gamma_i i_t + \gamma_\pi \pi_t$ (i=money-market rate, π=inflation).

Definition 7.2.

$$\Delta \log D_t = \underbrace{\phi\big(\log D_{t-1} - \log D_{t-1}^*\big)}_{\text{error correction}} + \sum_k \theta_k\, \Delta x_{t-k} + \varepsilon_t, \qquad \phi < 0. \tag{7.2}$$

Readables: **elasticity** γ_i (rate sensibilité); **reversion speed** ϕ; **half-life** $\ln(0.5)/\ln(1+\phi)$. Fit with ridge/EN linear (pure OLS = λ=0).

Honest SEs. $T \approx 96$ serially-dependent points $\Rightarrow$ OLS SEs wrong. Use **HAC (Newey–West)** for t-stats + **moving-block bootstrap over months** for the elasticity CI (Ch. 10).

Artifact. Synthetic $\hat{\phi} > 0 \Rightarrow$ no cointegration, undefined half-life. Reported as such; real data decides.

7.3 Role

convention = legibility floor; ECM = economics; hazard = challenger. All three scored on **realized cohort book run-off** MAE on a held-out window; winner = governed frozen choice (Ch. 11). Deploy the hazard only when it beats both OOT.

Chapter 8

Regimes: HMM and Stressed Survival

Smooth models smear discrete change; DZD attrition *steps* at devaluations / liquidity shifts. Add a regime layer in two roles.

8.1 Gaussian HMM

Definition 8.1. Latent $z_t \in \{1, \dots, K\}$, transition $A_{jk} = \mathbb{P}(z_t{=}k \mid z_{t-1}{=}j)$, emission $x_t \mid z_t{=}k \sim \mathcal{N}(\mu_k, \Sigma_k)$ (diagonal Σ). x_t = macro vector.

Three inferences ($\sim$ 80 lines stdlib):

- **filter** $\mathbb{P}(z_t{=}k \mid x_{1:t})$ — causal (forward recursion);
- **smooth** $\mathbb{P}(z_t{=}k \mid x_{1:T})$ — non-causal (forward–backward);
- **learn** (μ, Σ, A, π) — Baum–Welch (EM).

Causality: as a feature, use **filtered** only (smoothed leaks the future); params fit on dev, applied forward. Two build fixes: (i) **standardize emissions** (else oil ~ 65 swamps cpi $\sim 0.05 \to$ single-state collapse, degenerate $A \sim 10^{-229}$); (ii) **BIC-select** $K \in \{2, 3\}$.

8.2 Role 1: posterior as hazard feature

At τ, the filtered posterior $\mathbb{P}(z_\tau{=}k \mid x_{1:\tau})$ (a simplex point) enters (3.3) as `regime_p*`. Makes the *base* $S(t)$ regime-aware. Deployed iff **Gate B** (Ch. 11) lowers frozen-OOS NLL; on synthetic it was dropped (no incremental signal). Real data decides.

8.3 Online (dynamic) HMM

Static Baum–Welch = one A for all time — wrong when *dynamics* change. Track discounted sufficient statistics with forgetting ρ, from **filtered** responsibilities:

$$W_k \leftarrow \rho W_k + \gamma_t(k), \quad S_k^{(1)} \leftarrow \rho S_k^{(1)} + \gamma_t(k) x_t, \quad T_{jk} \leftarrow \rho T_{jk} + \xi_t(j,k), \tag{8.1}$$

re-derive (μ_k, Σ_k, A_t); effective memory $\approx 1/(1-\rho)$ (ρ=0.97 $\Rightarrow\sim$ 33 mo). Tracks a 0.95 $\to$ 0.60 persistence shift a static fit blurs to $\sim$ 0.76. Warm-start from batch BW; filtered-only $\Rightarrow$ look-ahead-safe. Daily advances the *filter* (frozen params); re-segmentation = refit.

8.4 Role 2: CTMC stressed survival

Pure-jump Markov chain (not a diffusion: σ unidentifiable on 120 mo administered-rate data). Exogenous states {liquid, stagnant, currency-stress}, generator Q, regime-conditional hazard intercepts α_k, init p_0:

$$S(t) = \mathbf{1}^\top \exp((Q - \operatorname{diag}\alpha)\, t)\, p_0 \tag{8.2}$$

(matrix-exp by scaling-and-squaring + truncated series). Q is **set, not learned** ($\sim$ 2–3 episodes $\Rightarrow$ unidentifiable): date regimes exogenously, estimate only α_k, treat Q sojourns as a scenario knob. Stress = shift $p_0 \to$ currency-stress / raise entry intensity $\Rightarrow$ degraded $S(t)$. Result = **scenario band, not coverage-certified** (Gate-A caveat).

Ablation. Data-driven hazard wins (book-$S(t)$ MAE $\approx$ 0.04); CTMC beats the constant floor ($0.12 < 0.17$) but is dominated. Stochastic toolbox = the stress generator, not the forecaster.

Chapter 9

Structural Breaks and Monitoring

Detect discrete change: retrospectively (where a single fit is illegitimate) and online (alarm on staleness). All causal — normalization by future data is the leak to avoid.

9.1 CUSUM (mean break)

Recursive residuals w_t (expanding-window 1-step errors), i.i.d. mean-0 under stability.

$$S_t = \frac{1}{\hat{\sigma}} \sum_{j=k+1}^{t} w_j, \qquad n^{-1/2} S_{\lfloor rn \rfloor} \xrightarrow{d} B(r). \tag{9.1}$$

A mean break drifts the residuals $\Rightarrow$ S_t crosses $\pm c\sqrt{n}(1+2r)$ (c = Brownian-crossing quantile). Production: macro-stream **break alarm** $\Rightarrow$ flags `recommend_early_refit` (alarm only). $\hat{\sigma}$ must be expanding-window (full-sample $\hat{\sigma}$ = the Step-3 leak).

9.2 ICSS + Sansó (variance breaks)

$C_t = \sum_{j \le t} a_j^2$, centered statistic $D_t = C_t/C_n - t/n$ (D_0=D_n=0):

$$\sqrt{\tfrac{n}{2}}\, \max_t |D_t| \xrightarrow{d} \sup_r |B^0(r)|, \quad B^0(r) = B(r) - rB(1). \tag{9.2}$$

Candidate at $\arg\max_t |D_t|$; iterate (find, split, recurse, refine). Classical ICSS i.i.d.-Gaussian $\Rightarrow$ **over-detects under GARCH**. **Sansó** κ_2 uses a dependence-robust (HAC) scaling $\Rightarrow$ FP rate $0.41 \to 0.06$ (bw scale ≈ 3).

9.3 SADF (explosive episode)

Backward-expanding/rolling ADF, supremum:

$$\mathrm{SADF} = \sup_{r_2} \mathrm{ADF}_{0,r_2}, \qquad \mathrm{GSADF} = \sup_{r_1,r_2} \mathrm{ADF}_{r_1,r_2}. \tag{9.3}$$

Right-tailed (bubble = root > 1): reject when ADF large. Backward-expanding sequence date-stamps (cross / un-cross a simulated CV) ⇒ causal “explosive” flag (deposit flight).

9.4 Use

- **Diagnostic:** a flag ⇒ a single fit over that span averages regimes ⇒ expanding/rolling refit + embargo so a label window does not straddle a break (Ch. 11).
- **Monitoring:** daily CUSUM alarm, monitoring-only; re-segmentation/refit is governed (frozen-artifact contract).
- **Pitfall:** detection lag is intrinsic (need post-break data to confirm); never peek forward. The alarm means “a break is *by now* apparent.”

Chapter 10

Uncertainty: Bootstrap, Conformal, MC Stress

Two error sources; headline the binding one.

- **(a) Cross-sectional (accounts).** Account-cluster bootstrap (block = account, carry all its rows). **Narrow** (N_{acct} large).
- **(b) Temporal (common shock).** Accounts share each month's macro $\Rightarrow$ not independent. Time-block bootstrap. **Wide, binding.**

Honesty rule. Headline = time-block; account bootstrap = the small idiosyncratic component, labelled. The account band drastically understates uncertainty.

10.1 Block bootstrap

Definition 10.1 (moving-block). Resample contiguous month-blocks of length $\ell \approx$ macro ACF length (or $= H$); concatenate to length T; refit; recompute $S(t)$.

Definition 10.2 (stationary, Politis–Romano). Geometric random block lengths $\Rightarrow$ exactly stationary resample (removes fixed-ℓ periodicity).

Limit. At $T \approx 120$, high persistence $\Rightarrow$ coverage caps ~ 0.88 (too few independent blocks). Fix = fixed-b HAC / bootstrap-t. Disclosed.

10.2 BCa

Naive percentile is biased under skew/variance-dependence (bounded survival). Adjust endpoints by bias-correction $\hat{z}_0$ and acceleration $\hat{a}$ (jackknife):

$$\alpha_{1,2} = \Phi\Big(\hat{z}_0 + \frac{\hat{z}_0 \pm z_{1-\alpha/2}}{1 - \hat{a}(\hat{z}_0 \pm z_{1-\alpha/2})}\Big). \tag{10.1}$$

Second-order accurate, transformation-respecting. Default for $S(t)$.

10.3 Conformal

Definition 10.3 (split conformal). Nonconformity score $|S_{\text{real}}(t) - \hat{S}(t)|$ on a calibration block; band = empirical $(1-\alpha)$ quantile. Marginal coverage guaranteed under **exchangeability** (assumption-free distributionally).

Time series breaks exchangeability $\Rightarrow$ **weighted conformal** (reweight calibration toward the test regime) and/or **ACI** (update the quantile online as coverage drifts); **CQR** on quantile hazards. Synthetic split-conformal coverage = 0.90 exact at all noise levels.

Combined band: conformal (prediction) $\oplus$ time-block bootstrap (parameter). Under a genuine OOD rate shock (Gate A) coverage is **not** guaranteed $\Rightarrow$ scenario band, stated.

10.4 Monte-Carlo stress

HMM is generative $\Rightarrow$ simulate regime+macro paths through frozen hazard+erosion:

$$\text{fan} = \{p_{05}, p_{50}, p_{95}\} \text{ of } S(t), \qquad \text{WAL tail} = \{p_{01}, p_{99}\}. \tag{10.2}$$

For baseline, $\pm$200bp, adverse-regime. Synthetic: +200bp WAL 11.23 $\to$ 11.13 mo, monotone bands.

Scope. MC = macro-path uncertainty only (bands tight $\sim \pm 2\%$ at 238 accounts). Parameter uncertainty (binding on 120 mo) = time-block bootstrap. Full band = coef-bootstrap $\times$ MC (documented next layer); current ships the macro-path fan.

Chapter 11

Validation

$S(t)$ forecasts the future decay of today's stock $\Rightarrow$ out-of-time only.

11.1 Protocols

- **Anchored walk-forward (headline).** Train $[\text{start}, o]$, score $(o, o+K]$, step o; report mean $\pm$ dispersion across origins, never one fold. Mimics deployment.
- **CPCV (secondary).** Dev $\rightarrow$ N=8 super-blocks ($\sim$ 12 mo), k=2 test $\Rightarrow$ $\binom{8}{2}$=28 paths; report dispersion only (less deployment-faithful than walk-forward).

11.2 Purge / embargo

Hazard label overlaps 1 mo; aggregated $S(t)$ overlaps H.

Definition 11.1. *Purge*: drop train rows whose label window $[\tau, \tau + H]$ overlaps a test block. *Embargo*: drop a buffer after it. Both are **time predicates on τ (real months), not row counts**; width = H for $S(t)$, 1 mo for hazard discrimination. The one place event-time changes the splitter.

11.3 Nested selection (headline never selects)

Using the held-out tail to *select* and *report* inflates the headline. Fix: train $\rightarrow$ validation $\rightarrow$ test:

- **train** (walk-forward): HP search (λ, α);
- **validation:** feature family (base / +regime / +signatures) by NLL **and** run-off model (convention/ECM/hazard) by realized-cohort MAE;
- **test:** touched once = honest headline; never selects.

Two-level: within family L1 selects per-coef; across families validation NLL decides. **Parsimony tie-break** (1-SE spirit, family tol $\sim$ 0.01, model tol $\sim$ 0.05): deploy the simplest within tol. Demo: base+regime beat base by 0.3% (noise) $\Rightarrow$ parsimony drops regime, deploys base; signatures rejected (val 0.043 vs 0.026); model = hazard (test MAE 0.012 vs ECM 0.044 vs conv 0.055); honest test NLL ≈ 0.033, ECE ≈ 0.0036.

11.4 Gate B

Same OOT protocol decides each option (depth 3 vs 2, sig vs raw, regime on/off, hazard vs conv/ECM); keep the simplest OOT winner, deflate for the search. Every option re-tested each recalibration, never permanently parked.

11.5 Calibration scorecard

$S(t) = \prod(1-h) \Rightarrow$ level-sensitive. Discrimination (time-dependent AUC, Uno C, IPCW) secondary. Primary:

- **reliability** (predicted vs realized h by bin; also balance-weighted);
- **PIT** $\approx$ Uniform$(0,1)$, KS;
- **aggregate-$S(t)$ error** on the frozen tail;
- **coverage** — does the 90% time-block fan cover realized? (headline: a band that misses is worse than none).

Censoring administrative & non-informative $\Rightarrow$ IPCW justified-omitted (stated).

11.6 Validation file

Stability across windows/segments; **event-definition** sensitivity (re-fit under alt floor/dormancy rules); conv-vs-ECM-vs-hazard on identical origins; predicted-vs-realized WAL/duration; Tier 1–3 explainability (ECM elasticity/reversion; group-Lasso channel selection + ablation; PDP/ALE; signature depth attribution); honest limits (Gate A, horizon-vs-history, bootstrap cap, signature trade). That file — not one number — is the sign-off artifact.

Chapter 12

System, Governance, Limits

12.1 Frozen-artifact contract

Only recalibration writes coefficients; the daily scorer only reads.

$$\texttt{eval} \to \texttt{hp\_selected.json} \ \to \ \texttt{fit} \to \texttt{model.json} \ \to \ \texttt{daily} \to S(t).$$

Daily can run unattended: a noisy month cannot re-train. CUSUM may *recommend* a refit; refit is governed.

12.2 Two cadences

- **Routine (monthly):** re-fit *coefficients* on **all** data (no held-out tail — run-off must be current), frozen HPs. Fast, low-governance.
- **Governed (quarterly):** re-*select* HPs + family + model class (nested, Ch. 11); `model.json` flagged "pending MR sign-off". Heavier.

`eval` (reporting, held-out, **not** deployed) and `fit` (deployed, all-data, **not** the reported number) are separate entry points — conflating them = test-on-train.

12.3 Operational surface

One button:

```
python run_pipeline.py --dav-dir <dumps>               # monthly
python run_pipeline.py --dav-dir <dumps> --recalibrate # quarterly
```

Steps: `download` (macro fetch + panel rebuild), `eval`, `fit`, `daily` ($B(t)$), `stress` (MC fan + WAL tail), `report` (self-contained HTML). Plots = hand-written **SVG** (no matplotlib). Only aggregates leave the bank (coefs, curves, elasticities, calibration) — never client rows.

12.4 Explainability (governance)

Linear-on-features $\Rightarrow$ no SHAP needed. **T1** anchors (ECM elasticity/reversion; raw-feature logistic; monotone seasoning). **T2** group-Lasso channel selection + channel ablation (SHAP substitute) + PDP/ALE. **T3** signature depth attribution (functional-level only $\Rightarrow$ corroborate by ablation).

12.5 Spine (one line)

PIT event-time panel $\rightarrow$ Gate A $\rightarrow$ convention + HAC/block-bootstrap ECM floor $\rightarrow$ pooled balance-weighted calibrated penalized-logistic hazard on raw+signatures, with regime posterior + erosion $\rightarrow$ walk-forward (CPCV second) + purge/embargo + nested selection $\rightarrow$ balance-weighted $S(t), B(t)$ + time-block + conformal bands $\rightarrow$ regime overlay (HMM feature; CTMC stressed $S(t)$) $\rightarrow$ stress with **coverage** as headline.

12.6 Honest limits (not deployment-certified)

1. All numbers = methodology on synthetic real-format data; real panel on bank PC; first run needs `profile_dav.py` to confirm grain/scope/floor.
2. Frozen OOS 2023–24 single-regime $\Rightarrow$ regime-stress $S(t)$ fit-only.
3. Rate sensitivity weakly identified (Gate A) $\Rightarrow$ scenario-driven.
4. Bootstrap coverage caps ~ 0.88 at T=120 $\Rightarrow$ fixed-b HAC / bootstrap-t.
5. Coef-bootstrap $\times$ MC combine = named next layer; current fan is macro-path only.

Cadence = **monthly**, two jobs (frozen scoring vs governed recalibration); never daily retraining. The deliverable is the measured-vs-assumed boundary, not a single accuracy number.